Satya Phanindra Kalaga

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Research Interests

- Reliability and reproducibility of language model systems
- Accountability and auditability of autonomous agents
- AI policy and governance in education and the workforce

Education

Master of Science in Data Science Jan 2024–Dec 2025
George Washington University, Washington, D.C.

- GPA 3.90
- Relevant Coursework: Machine Learning, NLP, Data Mining, Computer Vision, Cloud Computing

Bachelor of Technology in Computer Science and Engineering Aug 2019–May 2023
Jawaharlal Nehru Technological University, India

Publications

Agent Security Meets Regulatory Reality: A Practitioner Systematization of Autonomous-Agent Threats and Controls in Regulated Financial Systems 2026
Krishna Mohan Guda Nagavenkata Srinivasa, **Satya Phanindra Kalaga**, and Karthik Guda Vishnu. *Software: Practice and Experience* (Wiley), submitted.
Contribution: revised the algorithms and several sections, and built the open-source reference implementation.
Code artifact

LAiSER: A Taxonomy-Aware Framework for Skill Extraction and Research Nov 2025
Satya Phanindra Kumar Kalaga, Bharat Khandelwal, Anket Vilasrao Patil, and Prudhviraju Chekuri. *Journal of Open Source Software*, under review.
JOSS submission and open peer review

Research Experience

Data Science Consultant, Open Syllabus Project Aug 2026–Present
VitalSource Technologies

- **AI Policy Research**: Lead the study of how AI-use policies in academic syllabi have changed over time, owning the data and analysis work end to end toward a public dashboard for students, universities, and journalists.
- **Language Model Data Quality**: Turned 32.9M unstructured syllabi into a structured, publishable dataset for $66 against a $220 to $650 budget. Hash-based deduplication cut the language model workload by 84%, and four automated quality gates held output validity at 99.2% and run-to-run agreement at κ 0.87 to 0.95.
- **Corpus Expansion**: Added a new data source to the 33.5M-syllabus production corpus, loading 1.67M documents (1.54 TB) after profiling about 894M rows of crawl metadata and finding 1.58M cloud-hosted files the crawlers had missed. Built fault-tolerant ingestion across four cloud storage APIs that raised throughput 25 to 35x and cut a projected 56-day backlog to 2.6 days.
- **Pipeline Monitoring**: Found that 53% of crawl jobs marked successful returned no data, by archiving and analyzing 1,197 jobs from a pipeline that had no monitoring. Built health rules and automated alerts that now flag stalled crawls within 40 minutes instead of 30 days.

Data Research Specialist (AI/ML), LAiSER Apr 2024–Jul 2026
Program on Skills, Credentials and Workforce Policy, Institute of Public Policy, George Washington University, Washington, D.C.

- **LLM Pipeline Scaling**: Built and scaled LLM extraction pipelines across 1M+ records using vLLM; improved inference throughput 7x, cutting end-to-end processing time from 2 months to 8 days for empirical research

workflows.

- **LLM API Integration**: Integrated Gemini API into LAiSER for multi-model evaluation workflows; designed abstraction layers to support rapid experimentation across different LLM providers.
- **Retrieval Performance**: Implemented FAISS-based semantic search and optimized retrieval, delivering a consistent 30% speed improvement for retrieval-augmented research workflows.
- **Workforce Analysis**: Developed Neo4j and Plotly visualizations to identify skill gaps across 1,000+ courses and provided recommendations on two local training policies for the Texas Workforce Commission.
- **Open-source Engineering**: Primary maintainer of LAiSER GitHub organization (multiple repos) with a team of 7; drove architecture decisions, reusable libraries, and rapid iteration to support reproducible research. Received the 2025 GW OSPO open-source student award for LAiSER.
- **Release Quality & DevSecOps**: Led releases for 4+ major versions using GitHub Actions; added CodeQL and Dependabot for automated scanning and alerts; standardized Conventional Commits and Semantic Versioning for transparent collaboration.
- **Research Communication**: Worked with non-technical partners to explain design choices and trade-offs; supported $1M in grant-funded research with the Walmart Foundation and the Gates Foundation, applying NIST AI RMF for risk management.

Data Science Intern, Data Science for Sustainable Development Aug 2024–Dec 2024
George Washington University, Washington, D.C.

- **Web Platform Integration**: Led a team of 5 to design and deploy a web-based analytics platform (Next.js, Flask) on AWS EC2 and S3, integrating data pipelines and APIs for collaborative research workflows.
- **Workflow Automation**: Built agentic workflows to generate insights and operational summaries, improving partner collaboration efficiency by 30%.

Teaching and Workshops

Workshop Instructor, Multi-Agent Systems and Model Context Protocol Nov 2025
DevFest Annapolis

- Delivered a technical workshop and talk on Model Context Protocol (MCP), multi-agent systems, and agent communication mechanisms for 100+ students and faculty on November 15, 2025.

Graduate Assistant, Natural Language Processing for Data Science (DATS 6312) Aug 2025–Dec 2025
George Washington University, Washington, D.C.

- **Autograding**: Built a Python autograding toolkit that reduced grading turnaround from 3 days to 2 hours across 10 assignments, saving 100 TA hours.
- **Course Infrastructure**: Standardized the course environment on AWS EC2 with a CUDA and PyTorch image, using S3 for artifacts, IAM for access, and CloudWatch for monitoring.
- **Course Analytics**: Built instructor dashboards and learner survey tools to track assignment and activity patterns.

Workshop Instructor, AI and Serverless Deployment on Google Cloud Oct 2025
DC Startup and Tech Week

- Delivered a technical workshop for 60+ startup founders on building with AI, containers, and serverless deployments on Google Cloud.

Talks and Presentations

Dec 2026	GW Coders Lunch and Learn: Invited talk, "The Open-Source Edge: When Sharing Your Code Becomes Your Start-up Moat," scheduled for December 2, 2026
2026	Grace Hopper Celebration 2026: Talk, "The Open-Source Edge: Sharing Code as a Startup Moat," selected from 1,300+ proposals, and poster on LAiSER skill-extraction research accepted; the conference was canceled
Jun 2026	United Nations Open Source Week 2026, Community Day: Panelist; proposal selected from 700+ submissions
Jun 2026	University of Chicago AI+Science Summer School: Presented a poster on LAiSER

2026 GWU Open Source Conference: Delivered two lightning talks, "Chaos to Clarity: Standards for Open Source Growth" and "Open Source AI for Real-Time Danger Detection"

Dec 2025 Build with AI - Startup Day: Organized a summit for 200 attendees featuring C-suite executives and senior engineers; delivered a session on the impact of open-source software on AI and business strategy

Jul 2025 Badge Summit at CU Boulder: Hosted a table talk with 50 leaders in higher-education research on skills and credentials

2025 United Nations Open Source Week 2025: My Own Medic selected for presentation

May 2025 GW Innovation Fest: Showcased LAiSER on May 1, 2025

May 2025 GWU Open Source Conference: Delivered a lightning talk on open-source software best practices

Apr 2025 GW CCAS Poster Day: Showcased LAiSER on April 2, 2025

Awards

Jan 2026 GWU Open-Source Student Awards: Received award for open-source contributions (Danger Detection with ML)

2025 QS Reimagine Education Awards: Bronze Award, AI in Education (Higher Education Institutions), for LAiSER; selected from over 1,600 global submissions

Jan 2025 GWU Open-Source Student Awards: Received 3rd prize for best open-source software (LAiSER)

Selected Projects

Reinforcement Learning for Pseudo-Labeling (Capstone Project) 2025

- Built a custom Gymnasium pseudo-labeling environment with policy gradients on MNIST, and implemented RLGSSL (Heidari et al., 2024) with adaptive reward and loss components on CIFAR-10. Reached 45.55% best accuracy with 1,000 labels and analyzed the instability introduced by the adaptive rewards. Open-source on GitHub.
- **Tools**: *PyTorch (torchvision), Gymnasium, TensorBoard*

My Own Medic (Selected for United Nations Open-Source Week 2025) 2025

- Prototyped an open-source, chat-based health information assistant on a hosted language model, and demonstrated prompting with a structured patient record using mock data. Open-source on GitHub.
- **Tools**: *Python, PHP, JavaScript, Groq API, Qwen 2.5*

Benchmarking Database Architectures for Network Analytics 2024

- Reproducible Python benchmark of MySQL, MongoDB, and Neo4j on a synthetic road network dataset to deliver recommendations on which database performs best for each query type.
- **Tools**: *Python, MySQL, MongoDB, Neo4j (graph algorithms), SQL recursive CTEs, MongoDB aggregation/lookup, psutil (performance monitoring), Faker*

Global CO_2 Emissions Analysis Dashboard 2024

- Built a dashboard analyzing 223 years of CO_2 data across countries, sectors, and income groups to identify relation between historical emissions and renewable adoption priorities.
- **Tools**: *Tableau, Python, pandas, Plotly, Matplotlib*

Danger Detection with ML (GWU Open-Source Student Award, 2026) 2021

- Built a real-time computer vision prototype that detects firearms and fire in video streams and sends email alerts with captured images, averaging 25 FPS on CPU. Open-source on GitHub.
- **Tools**: *Python, OpenCV, YOLOv3 (Darknet), NumPy, argparse, SMTP (email notifications)*

Service and Leadership

Co-organizer and Tech Lead, Google Developer Groups DC May 2026–Present

President, Google Developer Groups On-Campus at GWU Sep 2024–Aug 2026

- Led a 300+ member student organization for AI/ML awareness through events and workshops; cohosted DevFest DC (900 attendees) and DevFest Annapolis (250+ attendees)

Chairperson of Relations, Data Science Association 2024–May 2025

- Cultivated relationships with C-suite executives and senior engineers for hosting events to provide career insights to 200+ data science students

Professional Experience

Full-Stack Engineer Jul 2021–Jan 2023

NashAgri, An Agritech B2C Organization, Maharashtra, India

- **Product Systems**: Built a cross-platform CRM from scratch to manage vendor transactions and inventory; added ML analytics to improve operational visibility and decision-making.
- **Data & Geo Analytics**: Managed geospatial data for 45,000+ farmers across 25,000+ regions (MongoDB, GeoJSON) to improve supply chain visibility and generate strategic insights.
- **Backend Features**: Delivered 5+ core features (invoice generation, RBAC, real-time auctions) supporting a 40% increase in vendor transactions in the first quarter.

Certifications

Google Cloud Professional Cloud Architect Jan 2026

Technical Skills

Languages: Python, SQL, JavaScript, R

ML & LLMs: NLP, LLMs, vLLM, PyTorch, scikit-learn, Hugging Face

Data, Cloud & MLOps: Pandas, NumPy, Spark, Airflow, Docker, AWS (EC2, S3, IAM, CloudWatch), GCP, Git, GitHub Actions, CI/CD, MLflow

Databases & Product: PostgreSQL, MySQL, MongoDB, Neo4j, FastAPI, Flask, Next.js, React, Plotly, Matplotlib, Tableau